Social Consideration in Simple and Complex Decision Problems

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Motivation

People's systematic deviation from the standard equilibrium prediction is common in experimental tests of decision theory.

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An important observation about the experimental game studies: when explaining systematic patterns of deviations,

- Others focus on **bounded rationality or bias**, e.g.,

$$a_i \tilde{\in} \arg\max_{a_i} U_i(a_i, a_{-i}),$$

  – Matching Problem: Hassidim et al.(2017); Auction Game: Ngangoué and Schotter (2023), Merlob et al. (2012), Ivanov et al. (2010); Repeated Game: Proto et al. (2019, 2022), Breitmoser (2015); Speculation Game: Pouget and Moinas (2013); Belief, Cognition and Decisions Experiments: Enke (2020), Taubinsky and ReesJones (2018), Altmann et al. (2022); Ali et al. (2021), Weizsäcker (2010) (2022); Error and Preference Experiments: Cason and Plott (2014), Nielsen and Rehbeck (2022); Others: Calford and Oprea (2017), Reiley et al. (2010)

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- Some focus on **other-regarding preferences**, e.g.,

$$\max_{a_i} \; U_i(a_i, a_{-i}) \; + \; \lambda \, U_{-i}(a_i, a_{-i}), \qquad \lambda \in [0, \bar{\lambda}].$$

  - Ultimatum Game: Andersen et al. (2011); Giving Game: Ottoni-Wilhelm et al. (2017), Landry et al. (2010), Kessler (2017), Erkal et al. (2011), Andreoni et al. (2017); Dictator Games: Di Tella et al. (2015), Brock, Lange, and Ozbay (2013); Reciprocal Game: Isoni et al. (2023), Castilla (2015); Gift Exchange Game: Malmendier and Schmidt (2017), Landry et al. (2010); Public Goods Game: Markussen et al. (2014); Minimum Effort Game: Friebel et al. (2017), Chen and Chen (2011); Other: Imas and Madarász (2024), Cappelen et al. (2013), Andreoni, et al. (2020)

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- Also, several discuss **both** sources, e.g., Bó et al. (2018), Hoppe and Schmitz (2013), Fehr et al. (2011, 2013), Gurdal et al. (2013), Enke and Zimmermann (2017), Cabrales et al. (2010)

Motivation

An important observation about the experimental game studies: when explaining systematic patterns of deviations,

- Others focus on **bounded rationality or bias**.
- Some focus on **other-regarding preferences**.
- Also, several discuss **both** sources.

But, why?

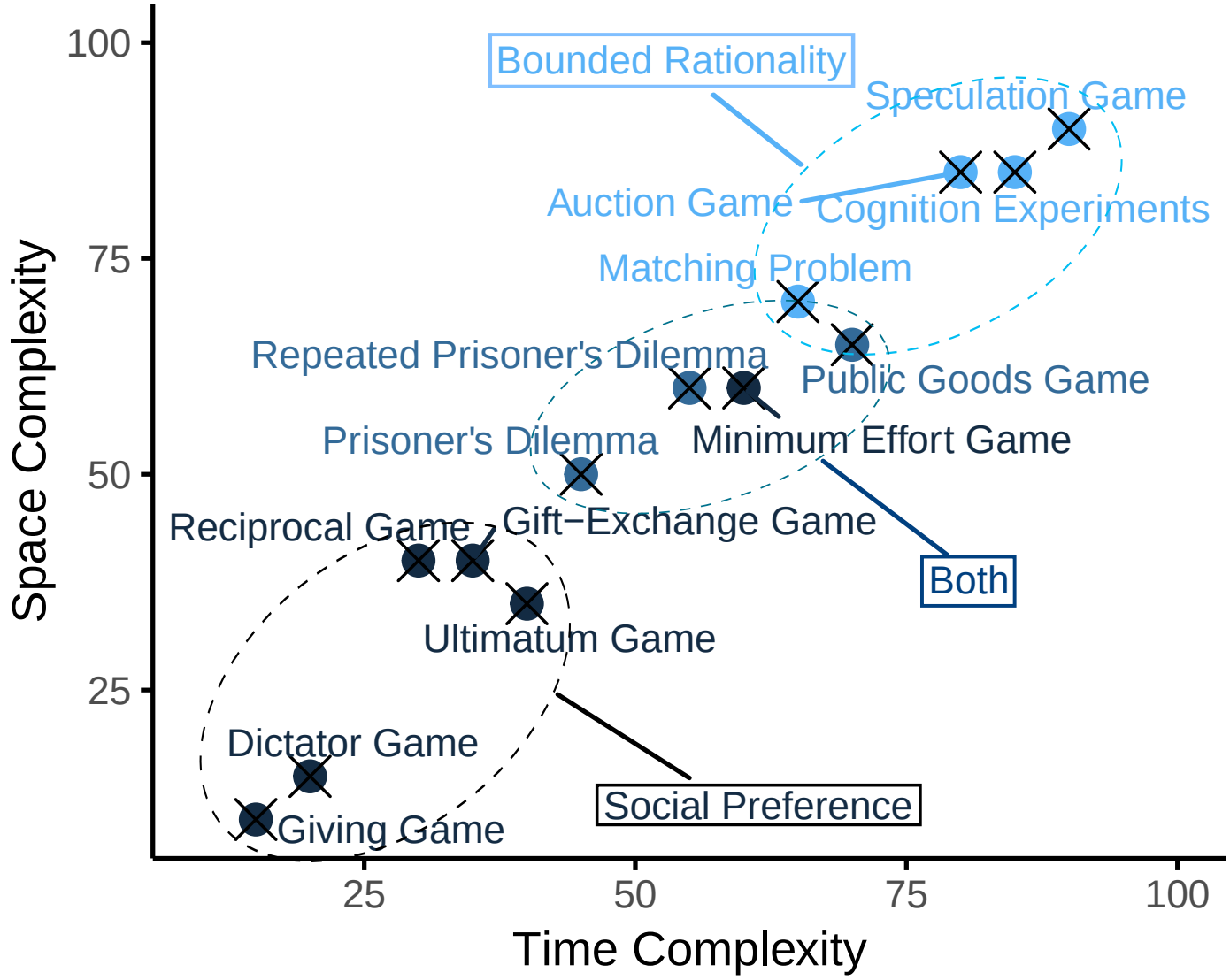
Motivation

> *...one fundamental question that...how one can reconcile the existence of social preferences with evidence from other paradigmatic experiments that appear to indicate that people are largely selfish...*
>
> — Fehr & Charness (2025)

- **Is there a fundamental factor that distinguishes these two types deviations?**

- Intuitively, it seems that games involving discussion of social preferences are **less complex** than those games involving discussion of bounded rationality.

A tentative preliminary attempt based on LLM



It shows that: Complexity might be an important criterion for identifying why predictions fail!

Research Questions

As the **complexity** level of a decision increases, will **other-regarding preferences** of playing the decision be eroded?

Our Answer

1. A Rational-choice Framework

We develop a *rational-choice framework* that can explain why behavior might appear selfish in some contexts while pro-social in others, even if the underlying social preferences themselves are stable.

- **Expression of social preferences (Active social consideration) is endogenous to the complexity of the decision problem.**

Key intuition: activating and reasoning with social considerations is cognitively costly, and agents *rationally* choose not to engage in it when the environment becomes too complex.

2. Experimental Evidence (In Progress)

We use a **computation-based treatment** to provide a simple and clean experimental test of the theoretical framework.

Two-Stage Procedural Decision Making

Standard approach

Agent i maximizes: $u_i(\lambda_i, s_i, s_{-i}) := u_i^0(s_i, s_{-i}) + \lambda_i u_i^1(s_i, s_{-i})$.

- strategies of i and -i: s_i, s_{-i}.
- material payoff and social payoff: $u_i^0(s_i, s_{-i})$, $u_i^1(s_i, s_{-i})$.
- weight assigned to the social payoff: $\lambda_i \geq 0$.

Two-stage procedural decision making (this project)

Stage 1: Deciding your procedures' informativeness.

Stage 2: Given the objective function representation under the informativeness, Do the rational decision.

Two-Stage Procedure: Informativeness Choice

Standard approach

Agent i maximizes: $u_i(\lambda_i, s_i, s_{-i}) := u_i^0(s_i, s_{-i}) + \lambda_i u_i^1(s_i, s_{-i})$.

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- weight assigned to the social payoff: $\lambda_i \geq 0$.

Two-stage procedural decision making (this project)

In the **first stage**:

$$\max_{(m_{i0}, m_{i1}) \in [0,1] \times [0,1]} v_i(m_{i0}, m_{i1}; m_{-i0}, m_{-i1}) - C_i(m_{i0}, m_{i1}, \kappa).$$

The agent first selects a **decision procedure** $p \in \mathcal{P}$, formalized by an **informativeness** vector $(m_0, m_1) \in [0, 1] \times [0, 1]$, to mentally represent the material and social payoff functions.

- (m_0, m_1): precision of the agent's Bayesian inference regarding the material and social payoff functions.
- Cognitive costs are incurred.

Model Primitives: Material Informativeness m_0

For example,

$$\max\ U(\text{material payoff},\ \text{social payoff},\ \text{interaction rule})$$

1. m_0: The informativeness we need for a problem representation without prosociality

The agent learns their and their opponent's material payoff functions through deliberation, modeled as acquiring an informative mental signal.

- The higher m_0 is, the more informative the mental signal is to reveal the material-related payoff functions;
- When $m_0 = 1$, the signal is perfectly revealing; When $m_0 = 0$, the signal is not informative.

Model Primitives: Social Informativeness m_1

For example,

$$\max\ U(\text{material payoff},\ \text{social payoff},\ \text{interaction rule})$$

1. m_0: The informativeness we need for a problem representation without prosociality

2. m_1: The additional informativeness we need for a problem representation with prosociality

The agent learns their social payoff functions by evaluating their material payoffs relative to others, modeled as acquiring a *distinct* informative mental signal.

- The higher m_1 is, the more informative the mental signal is to reveal the social payoff functions.

Model Primitives: Complexity κ & Default Procedure p^d

1. Complexity κ : Resources we need to be a rational agent

Definition

The complexity of a decision problem Γ *is measured by* $\kappa \in [0, \infty)$*, defined as the objective resource costs required to implement the rational procedure* p^r*.*

2. Default Procedure p^d: what we do when we know nothing

Definition

In a decision problem Γ*, let* p^d *be the default procedure with* $(m_0^d, m_1^d) = (0, 0)$*.*

Two-Stage Procedure: Rational Choice

Standard approach

Agent i maximizes: $u_i(\lambda_i, s_i, s_{-i}) := u_i^0(s_i, s_{-i}) + \lambda_i u_i^1(s_i, s_{-i})$.

- strategies of i and -i: s_i, s_{-i}.
- material payoff and social payoff: $u_i^0(s_i, s_{-i}), u_i^1(s_i, s_{-i})$.
- weight assigned to the social payoff: $\lambda_i \geq 0$.

Two-stage procedural decision making(this project):

The Second-stage

The second-stage objective is captured by the value function $v_i(m_{i0}, m_{i1}; m_{-i0}, m_{-i1})$:

$$v_i(m_{i0}, m_{i1}; m_{-i0}, m_{-i1}) := \max_{s_i \in \mathcal{S}_i} \mathbb{E}_{s_{-i}} \left[u_i(\lambda_i, s_i, s_{-i}) \mid m_{i0}, m_{i1}; m_{-i0}, m_{-i1} \right],$$

The agent then selects a strategy to maximize the mentally represented payoff function.

Stage 1:

Informativeness Choice

Stage 1: Assumptions

Technical Assumption. $v(m_0, m_1)$ and $C(m_0, m_1, \kappa)$ are twice differentiable.

Assumption 1

$\frac{\partial^2 C}{\partial m_0 \partial \kappa} > 0$, *and* $\frac{\partial^2 C}{\partial m_1 \partial \kappa} > 0$.

- The marginal cost of both m_0 and m_1 is increasing in decision problem complexity.

Assumption 2

$\frac{\partial^2 (v-C)}{\partial m_0 \partial m_1} \geq 0$.

- The objective in the first stage $v - C$ is supermodular in (m_0, m_1): m_0 and m_1 are *complementary* or *independent* in terms of their (marginal) impacts on utility.

Assumption 3

Let $x = (m_0, m_1)$. $\lim_{\kappa \to \infty} [\frac{\partial C}{\partial x_j}(x; \kappa) - \frac{\partial v}{\partial x_j}(x)] > 0, j = 0, 1$.

- The marginal cost exceeds the marginal reward as the decision problem becomes overwhelmingly complex.

Stage 1: Individual Decision-making – Comparative Statics

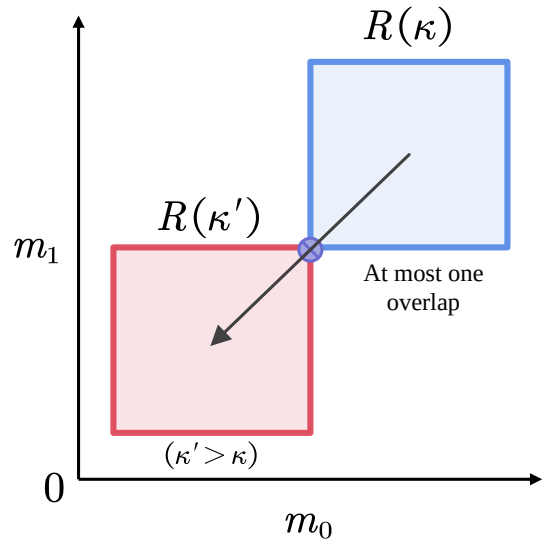
The optimal choice set: $R(\kappa) := \arg\max_{(m_0,m_1)\in[0,1]\times[0,1]} v(m_0, m_1) - C(m_0, m_1, \kappa)$.

Proposition 1 (Optimal informativeness Choice Non-increasing in Complexity)

Under Assumptions 1 and 2,

(1) $R(\kappa)$ *is nonempty, and is closed under the operation of* $\wedge$ *and* $\vee$*, where* $x \wedge x' := (min(x_1, x_1'), min(x_2, x_2'))$ *and* $x \vee x' := (max(x_1, x_1'), max(x_2, x_2'))$.

(2) *For any* $(m_0, m_1) \in R(\kappa)$ *and* $(m_0', m_1') \in R(\kappa')$ *such that* $\kappa' > \kappa$*, it follows that* $(m_0, m_1) \geq (m_0', m_1')$.



Comparative Statics

Proposition 1

[Proof sketch]

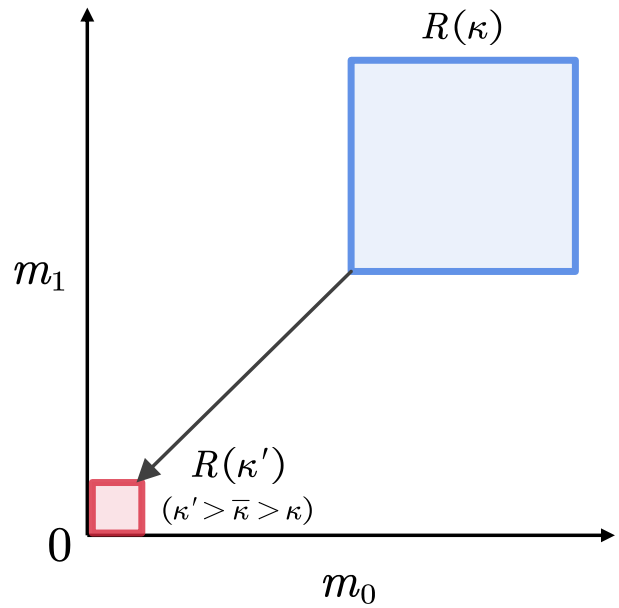
Stage 1: Individual Decision-making – Complete Collapse

Proposition 2 (Convert to Default and Social Considerations Vanish)

Under Assumptions 1 and 3:

$$\exists\, \bar{\kappa} < \infty,\ \forall\, \kappa > \bar{\kappa},$$
$$R(\kappa) := \arg\max_{(m_0, m_1)} v(m_0, m_1) - C(m_0, m_1, \kappa) = (0, 0).$$

- When a problem becomes sufficiently complex, the agent will eventually give up on all deliberation including active social consideration.



Complete Collapse

Proposition 2

[Proof sketch]

Stage 1: Strategic Games – Additional Assumption

Assumption 4

$\frac{\partial^2 v_i}{\partial m_{ij} \partial m_{-is}} \geq 0$, *where* $j, s \in \{0, 1\}$.

- Agents' choices of informativeness of mental signals exhibit strategic *complementarity* or *independence*.
- If opponents start to employ more thoughtful, fine-tuned procedures or care more about others or social outcomes, it is beneficial or at least not harmful for the agent to do the same.

Stage 1: Game Equilibrium – Comparative Statics

$\mathcal{E}(\kappa)$: set of Nash equilibria in game Γ with complexity level κ.

Reminder:

Assumption 1: $\frac{\partial^2 C}{\partial m_0 \partial \kappa} > 0$, and $\frac{\partial^2 C}{\partial m_1 \partial \kappa} > 0$.

Assumption 2: $\frac{\partial^2 (v-C)}{\partial m_0 \partial m_1} \geq 0$.

Assumption 4: $\frac{\partial^2 v_i}{\partial m_{ij} \partial m_{-is}} \geq 0$, where $j, s \in \{0, 1\}$.

Proposition 3 (Equilibrium Choice Non-increasing in Complexity)

Under Assumptions 1, 2, and 4:

(1) $\mathcal{E}(\kappa) \neq \emptyset$ *is a complete sublattice of lattice* $(X, \leq)$, *where* $X := \prod_{i=1}^{N} X_i$. *Both* $\underline{x}(\kappa) := \inf \mathcal{E}(\kappa)$ *and* $\overline{x}(\kappa) := \sup \mathcal{E}(\kappa)$ *exist.*

(2) *For* $\kappa' \geq \kappa$, $\underline{x}(\kappa') \leq \underline{x}(\kappa)$ *and* $\overline{x}(\kappa') \leq \overline{x}(\kappa)$.

- As game complexity increases, the set of equilibria shifts "southwest": both *minimum* amd *maximum* equilibrium level of material payoff and social payoff considerations weakly decrease in complexity.

[Proof sketch]

Stage 1: Game Equilibrium – Complete Collapse

Reminder:

Assumption 1: $\frac{\partial^2 C}{\partial m \partial \kappa} \geq 0$, and $\frac{\partial^2 C}{\partial \lambda \partial \kappa} \geq 0$.

Assumption 3: Let $x = (m_0, m_1)$. $\lim_{\kappa \to \infty} [\frac{\partial C}{\partial x_j}(x; \kappa) - \frac{\partial v}{\partial x_j}(x)] > 0, j = 0, 1$.

Proposition 4 (Convert to Non-deliberate Equilibrium)

Under Assumptions 1 and 3, $\exists\ \bar{\kappa} < \infty$, $\forall\ \kappa > \bar{\kappa}$, $\mathcal{E}(\kappa) = \{\mathbf{0}\}$.

- When game complexity exceeds a finite threshold, all strategic deliberation collapses simultaneously across players – convert to default procedure and social considerations vanish entirely.

[Proof sketch]

Stage 2:

Rational Choice Given Informativeness

Stage 2: Gaussian Information and Posterior Beliefs

Setup. Fix an individual i, an action $s \in \mathcal{S}$, and complexity κ. The underlying objective is $\Pi_i(s) + \bar{\lambda} F_s$, with $\bar{\lambda} > 0$. Assume action-invariant Gaussian prior means and a common prior variance:

$$\Pi_i(s) \sim \mathcal{N}(\Pi_0, \tau^2), \qquad F_s \sim \mathcal{N}(F_0, \tau^2), \qquad \tau^2 > 0.$$

Gaussian signals. For $m_0, m_1 \in [0,1]$, with positive signal precisions,

$$\begin{aligned} y_\Pi(s) \mid \Pi_i(s) &\sim \mathcal{N}\left(\Pi_i(s), \frac{\sigma^2}{m_0(\kappa)}\right), \\ y_F(s) \mid F_s &\sim \mathcal{N}\left(F_s, \frac{\sigma_F^2}{g(m_0(\kappa), m_1(\kappa))}\right), \end{aligned} \qquad \sigma_F^2 \geq \sigma^2 > 0.$$

One Bayesian update. Write

$$\alpha_\Pi = \frac{m_0}{\sigma^2}, \quad \alpha_F = \frac{g(m_0, m_1)}{\sigma_F^2}, \qquad \omega_j = \frac{\alpha_j}{\tau^{-2} + \alpha_j} \ (j \in \{\Pi, F\}).$$

Then the posterior expectations used for choice are

$$\begin{aligned} \mathbb{E}[\Pi_i(s) \mid y_\Pi(s)] &= (1 - \omega_\Pi)\Pi_0 + \omega_\Pi y_\Pi(s), \\ \mathbb{E}[F_s \mid y_F(s)] &= (1 - \omega_F)F_0 + \omega_F y_F(s). \end{aligned}$$

Stage 2: Posterior Optimization and Effective Social Weight

Individual forward decision. After observing the realized signals, choose

$$\max_{s\in\mathcal{S}} \left\{\mathbb{E}[\Pi_i(s) \mid y_\Pi(s)] + \bar{\lambda}\,\mathbb{E}[F_s \mid y_F(s)]\right\}.$$

Substituting the posterior means yields

$$\max_{s\in\mathcal{S}} \left\{(1-\omega_\Pi)\Pi_0 + \bar{\lambda}(1-\omega_F)F_0 + \omega_\Pi y_\Pi(s) + \bar{\lambda}\omega_F y_F(s)\right\}.$$

Conditions for the comparative statics.

- **Social information retention:** $g(m_0, m_1) = m_0 q(m_0, m_1), \quad 0 \le q \le 1$, $\partial q/\partial m_0 \ge 0$, $\partial q/\partial m_1 \ge 0$.
- **Cognitive monotonicity:** $\kappa' \ge \kappa \;\Rightarrow\; m_j(\kappa') \le m_j(\kappa)$ for $j = 0, 1$.

Proposition 5 (Effective Social Weight Is Non-increasing in Complexity)

For $m_0(\kappa) > 0$, removing the action-independent prior terms and normalizing by $\omega_\Pi > 0$ gives the equivalent choice problem

$$\max_{s\in\mathcal{S}} \left\{y_\Pi(s) + W_{\text{rel}}(\kappa) y_F(s)\right\}, \qquad W_{\text{rel}}(\kappa) = \bar{\lambda}\frac{\omega_F(\kappa)}{\omega_\Pi(\kappa)}.$$

For $\kappa' \ge \kappa$ with $m_0(\kappa'), m_0(\kappa) > 0$, $W_{\text{rel}}(\kappa') \le W_{\text{rel}}(\kappa)$.

[Proof]

Stage 2: Benchmark Prosocial Choice and Conditions

Benchmark. Each of $N < \infty$ individuals chooses between prosocial s_P and non-prosocial s_N, where

$$\Delta\Pi_i := \Pi_i(s_P) - \Pi_i(s_N) < 0, \qquad \Delta F_i := F_{s_P} - F_{s_N} \geq 0.$$

Observed choices. Conditional on actual payoffs, Gaussian signal errors are independent across the two actions and the two signal types. With $W_i = \bar{\lambda}\omega_{F,i}/\omega_{\Pi,i}$, define

$$D_i(\kappa) = \Delta\Pi_i + W_i(\kappa)\Delta F_i, \qquad R_i(\kappa) = \frac{2\sigma^2}{m_{0,i}} + \frac{2\sigma_F^2 W_i(\kappa)^2}{m_{0,i}q_i},$$

$$S_N^{\text{obs}}(\kappa) = \frac{1}{N}\sum_{i=1}^{N} \Pr(s_P \text{ chosen} \mid \Pi, F) = \frac{1}{N}\sum_{i=1}^{N} \Phi\left(\frac{D_i(\kappa)}{\sqrt{R_i(\kappa)}}\right),$$

where $q_i = q(m_{0,i}, m_{1,i})$ and Φ is the standard normal CDF.

Additional conditions for the interior-dip result.

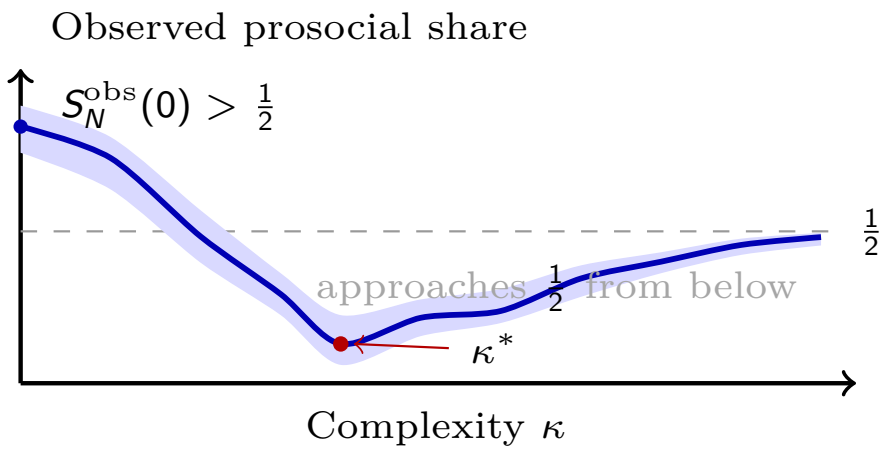
- At every finite κ: $m_{0,i} > 0$, $q_i > 0$, with continuous and interior-differentiable informativeness paths (and the preceding Gaussian setup).
- **(A) Cognitive exhaustion:** $m_{0,i}(0) = m_{1,i}(0) = 1$ and $\lim_{\kappa\to\infty} m_{0,i}(\kappa) = \lim_{\kappa\to\infty} m_{1,i}(\kappa) = 0$.
- **(E′) Retention boundary:** $q(0,0) = 0$ and q is continuous at $(0,0)$.
- **Initial condition:** $S_N^{\text{obs}}(0) > \frac{1}{2}$.

Stage 2: An Interior Dip in Observed Prosocial Choice

Proposition 6 (Interior Prosocial Low Point and Return to 1/2)

Under the preceding Gaussian benchmark and conditions, some finite $\kappa^* \in (0, \infty)$ *satisfies*

$$S_N^{\text{obs}}(\kappa^*) < S_N^{\text{obs}}(0), \qquad S_N^{\text{obs}}(\kappa^*) < \lim_{\kappa \to \infty} S_N^{\text{obs}}(\kappa) = \frac{1}{2}.$$



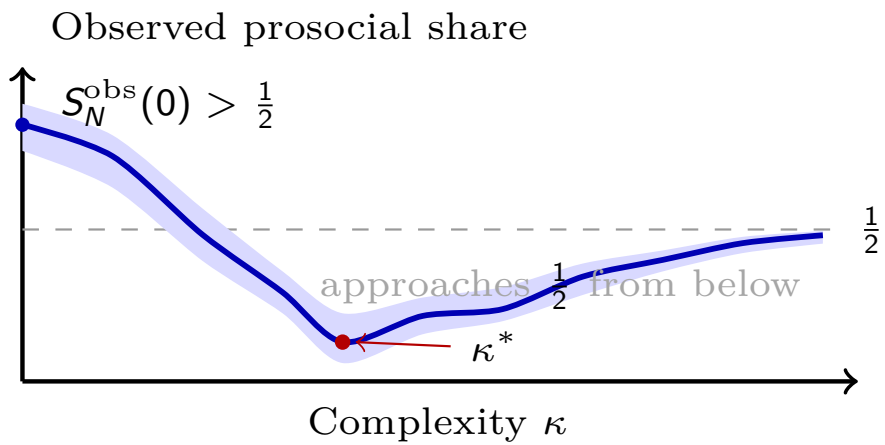
Illustrative path: return to 1/2 from below; no unique or strict U-shape implied.

[Proof]

Stage 2: Two Forces Behind the U-Shaped Tendency

Two opposing effects of increasing complexity:

- **Erosion of effective social consideration:** $W_{\mathrm{rel},i}(\kappa)$ falls, weakening the social term relative to the private-payoff sacrifice ($\Delta\Pi_i < 0$). This pushes prosocial choice down.
- **Increasing cognitive noise:** As $m_{0,i}(\kappa) \to 0$, the Gaussian comparison variance $R_i(\kappa) \to \infty$. Choice becomes less precise and the observed prosocial share returns toward 50%.



Competing forces can generate an interior dip; a unique, globally strict U-shape is not required.

[Proof]

Experiment

In Progress: Experimental Test

Overall Design.

游戏说明

游戏规则

在每轮游戏中，玩家 A 在两个分配方案中选择一个，该选择决定本轮两名玩家各自的收益。玩家 B 不需要作出分配决策。

2	A 和 B
个方案	两位玩家角色

决策内容

玩家 A

玩家 A 的正式决策页面中会显示一个类似下面的两个分配方案，x,y,m,n均代表具体的数值，其中某些数值可能会以算术式的方式呈现而非直接给出。

方案	玩家 A 的收益	玩家 B 的收益
方案 1	*x* 个实验币	*y* 个实验币
方案 2	*m* 个实验币	*n* 个实验币

玩家 A 需要选择其中一个方案。玩家 A 选择的方案同时决定本轮游戏中玩家 A 和玩家 B 的收益。

玩家 B

玩家 B 无需作出任何决策，其收益由玩家 A 的选择决定。

In Progress: Experimental Test

Simple (Baseline).

你的决定

你在本场游戏中的角色为：玩家 A

请在以下两个方案中选择一个。你的选择将决定你（玩家 A）和玩家 B 的收益。

选择	方案	玩家 A 的收益	玩家 B 的收益
○	方案 1	**97** 个实验币	**23** 个实验币
○	方案 2	**87** 个实验币	**73** 个实验币

确认提交

In Progress: Experimental Test

Complex(Treatment).

你的决定

你在本场游戏中的角色为：玩家 A

请在以下两个方案中选择一个。你的选择将决定你（玩家 A）和玩家 B 的收益。

选择	方案	玩家 A 的收益	玩家 B 的收益
○	方案 1	**166 + 215 − 284** 个实验币	**23** 个实验币
○	方案 2	**178 + 164 − 255** 个实验币	**73** 个实验币

确认提交

In Progress: Experimental Test

Super-Complex(Treatment).

你的决定

你在本场游戏中的角色为：玩家 A

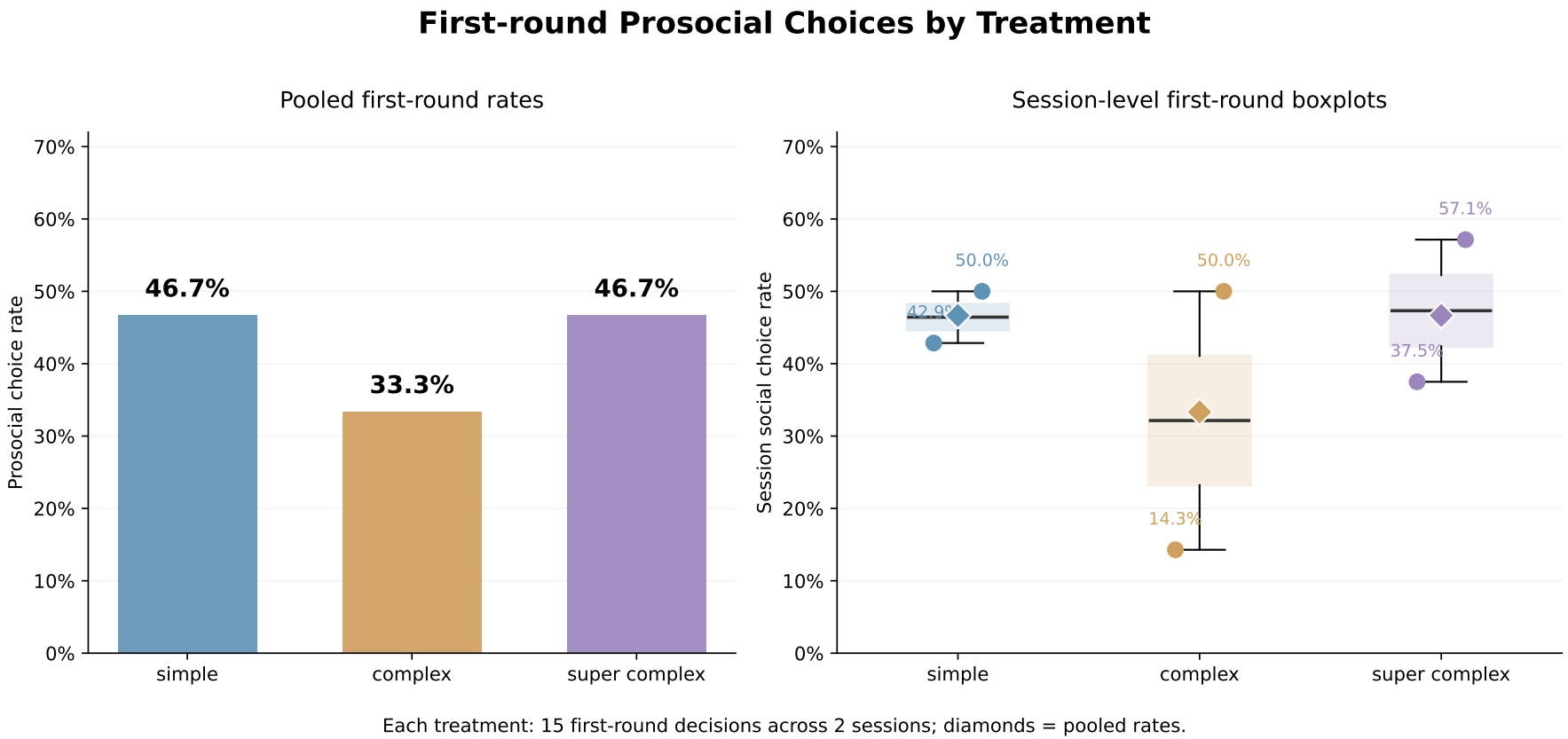
请在以下两个方案中选择一个。你的选择将决定你（玩家 A）和玩家 B 的收益。

选择	方案	玩家 A 的收益	玩家 B 的收益
○	方案 1	**295 + 113 − 287 + 148 − 172** 个实验币	**23** 个实验币
○	方案 2	**238 + 135 − 242 + 183 − 227** 个实验币	**73** 个实验币

确认提交

In Progress: Experimental Test

Results.



Appendix

Optimal Choice Non-increasing in Complexity: Proof

The optimal choice set: $R(\kappa) := \underset{(m,\lambda)\in\mathcal{M}\times\Lambda}{\arg\max}\ U(m,\lambda;\kappa)$

Proposition 1 (Comparative Statics)

Under Assumptions 1 and 2,

(1) $R(\kappa) \neq \emptyset$.

(2) For any $(m,\lambda) \in R(\kappa)$ and $(m',\lambda') \in R(\kappa')$ such that $\kappa' \geq \kappa$, $(m,\lambda) \vee (m',\lambda') \in R(\kappa)$ and $(m,\lambda) \wedge (m',\lambda') \in R(\kappa')$.

Proof.

Since $M \times \Lambda$ is compact and U is continuous, $R(\kappa)$ is non-empty by Weierstrass theorem. For any $x, x' \in R(\kappa)$, Assumption 2 ensures $x \vee x'$, $x \wedge x' \in R(\kappa)$.

Let $x := (m,\lambda)$ and $x' := (m',\lambda')$. Note that $x \vee x' \geq x$ and $x \wedge x' \leq x$. By Assumption 1, $U(x;\kappa) - U(x \vee x';\kappa) \leq U(x;\kappa') - U(x \vee x';\kappa')$. By Assumption 2, $U(x;\kappa') - U(x \vee x';\kappa') \leq U(x \wedge x';\kappa') - U(x';\kappa')$. Then,

$0 \leq U(x;\kappa) - U(x \vee x';\kappa) \leq U(x \wedge x';\kappa') - U(x';\kappa') \leq 0$, where the first and third inequality come from $x \in R(\kappa)$ and $x' \in R(\kappa')$, respectively. Thus, it must be that $x \vee x' \in R(\kappa)$ and $x \wedge x' \in R(\kappa')$. □

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Convert to Default and Social Consideration Vanish: Proof

Proposition 2 (Complexity Threshold)

Under Assumptions 1 and 3:

$$\exists \bar{\kappa} < \infty,\ \forall\, \kappa > \bar{\kappa},\ R(\kappa) := \underset{(m,\lambda)\in\mathcal{M}\times\Lambda}{\arg\max}\ U(m,\lambda;\kappa) = (\mathbf{0},0).$$

Proof.

Step 1: Marginal Costs Eventually Dominate. By Assumption 3, for any choice $x = (m,\lambda) \neq (\mathbf{0},0)$, the net marginal costs become strictly positive as complexity approaches infinity: $\lim_{\kappa\to\infty}\left[\frac{\partial C}{\partial m_t}(x;\kappa) - \frac{\partial \pi}{\partial m_t}(x)\right] > 0$, $\lim_{\kappa\to\infty}\left[\frac{\partial C}{\partial \lambda}(x;\kappa) - \frac{\partial \pi}{\partial \lambda}(x)\right] > 0$

Step 2: Uniform Bound via Dini's Theorem.

- By Assumption 1, the net marginal cost functions are non-decreasing in κ.
- $M \times \Lambda$ is compact, limit functions are continuous.
- Dini's theorem: convergence is uniform
- $\Rightarrow \exists \bar{\kappa} < \infty$ and $\varepsilon > 0$ such that $\forall \kappa > \bar{\kappa}$, $\forall x \in M \times \Lambda$: $\frac{\partial C}{\partial m_t} - \frac{\partial \pi}{\partial m_t} \geq \varepsilon$, $\frac{\partial C}{\partial \lambda} - \frac{\partial \pi}{\partial \lambda} \geq \varepsilon$

Step 3: Integration Along Any Path. For any $\kappa > \bar{\kappa}$ and any $x \neq (\mathbf{0},0)$:
$U(x;\kappa) - U(\mathbf{0},0;\kappa) = \int_{(\mathbf{0},0)}^{x}[\nabla\pi - \nabla C]\cdot ds \leq -\varepsilon\cdot(||m||_1 + \lambda) < 0$ □

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Equilibrium Choice Non-increasing in Complexity: Proof

$x_i := (m_i, \lambda_i)$ Let $B_i(x_{-i}; \kappa) := \arg\max_{x_i \in X_i} U_i(x_i, x_{-i}; \kappa)$, and $B_\kappa : X \rightrightarrows X$, with $B_\kappa(x) := \prod_{i=1}^N B_i(x_{-i}; \kappa)$ and $X := \prod_{i=1}^N X_i$.

Lemma 1 (Best Response Monotonicity in Complexity)

Let $\kappa' \geq \kappa$, $x_i' \in B_i(x_{-i}; \kappa')$, and $x_i \in B_i(x_{-i}; \kappa)$.
Under Assumptions 1 and 2: $x_i' \wedge x_i \in B_i(x_{-i}; \kappa')$ and $x_i' \vee x_i \in B_i(x_{-i}; \kappa)$.

Lemma 2 (Best Response Monotonicity in Opponents' Strategies)

Let $x'_{-i} \geq x_{-i}$, $x_i' \in B_i(x'_{-i}; \kappa)$, and $x_i \in B_i(x_{-i}; \kappa)$.
Under Assumptions 2 and 4: $x_i' \wedge x_i \in B_i(x_{-i}; \kappa)$ and $x_i' \vee x_i \in B_i(x'_{-i}; \kappa)$.

Lemma 3 (Properties of Best Response Correspondence)

Under Assumptions 1, 2, and 4, $B_\kappa : X \Rightarrow X$ satisfies:

- $B_\kappa(x)$ is non-empty, compact, and a sublattice of X for each x.
- The graph of B_κ is closed.
- If $x \leq y$, then for any $u \in B_\kappa(x)$, $v \in B_\kappa(y)$: $u \wedge v \in B_\kappa(x)$ and $u \vee v \in B_\kappa(y)$.

Lemma 4 (Extreme Best Responses)

Let $\underline{B}_\kappa(x) := \inf B_\kappa(x)$, $\bar{B}_\kappa(x) := \sup B_\kappa(x)$. Under Assumptions 1, 2, and 4:

- $\underline{B}_\kappa(\cdot)$ and $\bar{B}_\kappa(\cdot)$ are single-valued, non-decreasing, and continuous.
- They satisfy order-continuity properties for monotone sequences.

Equilibrium Choice Non-increasing in Complexity: Proof

Proposition 3 (Equilibrium Existence and Comparative Statics)

Under Assumptions 1, 2, and 4:

(1) $\mathcal{E}(\kappa) \neq \emptyset$ is a complete sublattice of $(X, \leq)$. Both $\underline{x}(\kappa) := \inf \mathcal{E}(\kappa)$ and $\bar{x}(\kappa) := \sup \mathcal{E}(\kappa)$ exist.

(2) For $\kappa' \geq \kappa$, $\underline{x}(\kappa') \leq \underline{x}(\kappa)$ and $\bar{x}(\kappa') \leq \bar{x}(\kappa)$.

Step 1: Existence and Sublattice Structure

- Let $B_i(x_{-i}; \kappa) := \arg\max_{x_i \in X_i} U_i(x_i, x_{-i}; \kappa)$, and $B_\kappa : X \rightrightarrows X$ with $B_\kappa(x) := \prod_{i=1}^N B_i(x_{-i}; \kappa)$ and $X := \prod_{i=1}^N X_i$.
- By Lemma 3, $B_\kappa(x)$ satisfies conditions for Tarski's fixed point theorem.
- Therefore $\mathcal{E}(\kappa) = \{x : x \in B_\kappa(x)\}$ is non-empty and forms a complete sublattice.

Step 2: Constructive Characterization

- Let $\bot$ and $\top$ be the least and largest element in X according to partial order $\leq$.
- Define iterations: $L^{(0)} = \bot$, $L^{(t+1)} = \underline{B}_\kappa(L^{(t)})$ and $U^{(0)} = \top$, $U^{(t+1)} = \bar{B}_\kappa(U^{(t)})$.
- By Lemma 4's monotonicity: $L^{(t)} \uparrow \underline{x}(\kappa)$ and $U^{(t)} \downarrow \bar{x}(\kappa)$.

Step 3: Comparative Statics

- Lemma 1 implies: $\kappa' > \kappa \Rightarrow B_{\kappa'}(x) \leq B_\kappa(x)$ for all x.
- By induction: $L^{(t)}(\kappa') \leq L^{(t)}(\kappa)$ and $U^{(t)}(\kappa') \leq U^{(t)}(\kappa)$.
- Taking limits: $\underline{x}(\kappa') \leq \underline{x}(\kappa)$ and $\bar{x}(\kappa') \leq \bar{x}(\kappa)$.

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Convert to Non-deliberative Equilibrium: Proof

Proposition 4 (Complexity Threshold in Games)

Under Assumptions 1 and 3, $\exists\, \bar{\kappa} < \infty$, $\forall\, \kappa > \bar{\kappa}$, $\mathcal{E}(\kappa) = \{\mathbf{0}\}$.

Step 1: Individual Player Thresholds

- For each player i, others' strategies only affect reward π_i, not costs C_i
- Similar to Proposition 2: $\exists \bar{\kappa}_i$ such that $\forall \kappa > \bar{\kappa}_i$: $B_i(x_{-i}; \kappa) = \mathbf{0}$ for all x_{-i}

Step 2: Uniform Convergence via Dini's Theorem

- By Assumption 3: $\lim_{\kappa \to \infty} [MC_i(x; \kappa) - MR_i(x)] = L_i(x) > 0$
- With Assumption 1, compact domain X + monotonicity $\Rightarrow$ uniform convergence
- $\exists \delta > 0$ such that $L_i(x) \geq \delta$ for all $x \in X$
- $\exists \bar{\kappa}_i$ such that for $\kappa > \bar{\kappa}_i$: $MC_i(x; \kappa) - MR_i(x) \geq \frac{\delta}{2}$

Step 3: Global Threshold

- Define $\bar{\kappa} = \max\{\bar{\kappa}_1, \ldots, \bar{\kappa}_N\}$
- For $\kappa > \bar{\kappa}$: Every player's unique best response is $\mathbf{0}$

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Proposition 5: Effective Social Weight – Proof

Step 1: Bayesian normalization. By the Gaussian posterior means and action-invariant prior means,

$$\arg\max_{s\in\mathcal{S}}\{\mathbb{E}[\Pi_i(s)\mid y_\Pi(s)]+\bar\lambda\mathbb{E}[F_s\mid y_F(s)]\}=\arg\max_{s\in\mathcal{S}}\{\omega_\Pi y_\Pi(s)+\bar\lambda\omega_F y_F(s)\}.$$

When $m_0>0$, division by $\omega_\Pi>0$ gives the normalized objective with $W_{\text{rel}}=\bar\lambda\omega_F/\omega_\Pi$.

Step 2: Comparative statics at the primitive level. Let $c=\tau^{-2}>0$, $\alpha=\alpha_\Pi=m_0/\sigma^2>0$, and

$$\rho=\frac{\alpha_F}{\alpha_\Pi}=\frac{\sigma^2}{\sigma_F^2}q(m_0,m_1)\in[0,1],\qquad \frac{W_{\text{rel}}}{\bar\lambda}=\Psi(\rho,\alpha):=\rho\frac{c+\alpha}{c+\rho\alpha}.$$

Its partial derivatives satisfy

$$\frac{\partial\Psi}{\partial\rho}=\frac{c(c+\alpha)}{(c+\rho\alpha)^2}>0,\qquad \frac{\partial\Psi}{\partial\alpha}=\frac{\rho c(1-\rho)}{(c+\rho\alpha)^2}\geq 0.$$

By m_0, m_1 non-increasing in κ and q non-decreasing in both arguments, ρ and α are non-increasing in κ. Therefore $W_{\text{rel}}(\kappa')\leq W_{\text{rel}}(\kappa)$ for any $\kappa'\geq\kappa$ with $m_0(\kappa'),m_0(\kappa)>0$.

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Proposition 6: Interior Prosocial Low Point – Proof (1/2)

For each individual i, abbreviate $m_i = m_{0,i}(\kappa)$ and $q_i = q(m_{0,i}(\kappa), m_{1,i}(\kappa))$. Write $a = \tau^{-2}\sigma^2 > 0$ and $b = \tau^{-2}\sigma_F^2 > 0$.

Step 1: Effective social weight vanishes. By (A), $(m_{0,i}, m_{1,i}) \to (0,0)$, and by (E′), $q_i \to 0$. Hence

$$W_i(\kappa) = \bar{\lambda}\frac{q_i(\kappa)[a + m_i(\kappa)]}{b + m_i(\kappa)q_i(\kappa)} \longrightarrow 0.$$

The deterministic signal-comparison margin consequently satisfies

$$D_i(\kappa) = \Delta\Pi_i + W_i(\kappa)\Delta F_i \longrightarrow \Delta\Pi_i < 0.$$

Since N is finite and $R_i(\kappa) > 0$, there exists $K < \infty$ such that for every i and $\kappa \geq K$,

$$\Pr(s_P \text{ chosen} \mid \Pi, F) = \Phi\left(\frac{D_i(\kappa)}{\sqrt{R_i(\kappa)}}\right) < \frac{1}{2}.$$

Therefore $S_N^{\text{obs}}(\kappa) < \frac{1}{2}$ for all $\kappa \geq K$.

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Proposition 6: Interior Prosocial Low Point – Proof (2/2)

Step 2: Cognitive uncertainty produces the 1/2 limit. For every finite κ, $m_i, q_i > 0$, and

$$R_i(\kappa) = \frac{2\sigma^2}{m_i(\kappa)} + \frac{2\sigma_F^2 W_i(\kappa)^2}{m_i(\kappa) q_i(\kappa)} \geq \frac{2\sigma^2}{m_i(\kappa)} \longrightarrow \infty.$$

Moreover $D_i(\kappa) \to \Delta\Pi_i < 0$, so $D_i(\kappa)/\sqrt{R_i(\kappa)} \to 0$. By continuity of Φ and finiteness of N,

$$\lim_{\kappa \to \infty} S_N^{\text{obs}}(\kappa) = \frac{1}{N} \sum_{i=1}^{N} \Phi(0) = \frac{1}{2},$$

with the limit approached from below for all sufficiently large finite κ.

Step 3: A finite interior global minimum exists. Choose $\bar{\kappa} \geq K$ and set $s_* = S_N^{\text{obs}}(\bar{\kappa}) < \frac{1}{2}$. The initial condition gives $S_N^{\text{obs}}(0) > \frac{1}{2} > s_*$. By the limit, choose a finite $T > \bar{\kappa}$ with

$$S_N^{\text{obs}}(\kappa) > \frac{s_* + 1/2}{2} > s_* \qquad \text{for all } \kappa \geq T.$$

Continuity and the extreme value theorem give a minimizer of S_N^{obs} on $[0, T]$; it is neither endpoint and is global on $[0, \infty)$. Therefore some finite $\kappa^* \in (0, \infty)$ obeys

$$S_N^{\text{obs}}(\kappa^*) \leq s_* < S_N^{\text{obs}}(0), \qquad S_N^{\text{obs}}(\kappa^*) < \frac{1}{2} = \lim_{\kappa \to \infty} S_N^{\text{obs}}(\kappa).$$

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